

CS257 Linear and Convex Optimization

Homework 2

Due: November 6, 2024

October 30, 2024

1. Let $f(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$ be an affine function from $\mathbb{R}^n$ to $\mathbb{R}^m$, where $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$. Show that if $C \subset \mathbb{R}^m$ is convex, so is its inverse image $f^{-1}(C) \triangleq \{\mathbf{x} : f(\mathbf{x}) \in C\}$.
2. Prove the convex hull $\text{conv } S$ of S is the set of all convex combinations of points in S by completing the following steps.

(a). Let

$$C = \left\{ \sum_{i=1}^m \theta_i \mathbf{x}_i : m \in \mathbb{N}; \mathbf{x}_i \in S, \theta_i \geq 0, i = 1, \dots, m; \sum_{i=1}^m \theta_i = 1 \right\}$$

Show that C is convex.

(b). Show that $C \subset \text{conv } S$ and conclude $C = \text{conv } S$.

3. Let $\mathbf{x}_0, \dots, \mathbf{x}_K$ be distinct points in $\mathbb{R}^n$. Let V be the set of points that are closer in Euclidean distance to $\mathbf{x}_0$ than $\mathbf{x}_1, \dots, \mathbf{x}_K$, called the **Voronoi region** around $\mathbf{x}_0$ with respect to $\mathbf{x}_1, \dots, \mathbf{x}_K$, i.e.

$$V = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_0\|_2 \leq \|\mathbf{x} - \mathbf{x}_i\|_2, i = 1, 2, \dots, K\}.$$

Show that V is a polyhedron by identifying $\mathbf{A}$ and $\mathbf{b}$ such that $V = \{\mathbf{x} : \mathbf{A}\mathbf{x} \leq \mathbf{b}\}$. (You don't have to draw it in the submission, but try to visualize V for $\mathbb{R}^2$.)

4. Determine if the following functions are convex, concave, or neither.

(a). $f(\mathbf{x}) = f(x_1, x_2, x_3) = x_1^2 + x_1x_3 + x_2^2 + x_2x_3 + \frac{1}{2}x_3^2$ on $\mathbb{R}^3$

(b). $f(\mathbf{x}) = f(x_1, x_2) = (x_1x_2)^{-1}$ on $\mathbb{R}_{++}^2 = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$

(c). $f(x_1, x_2) = x_1x_2^2$ on $\mathbb{R}_{++}^2 = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$

(d). $f(x_1, x_2) = x_1x_2^{-1/2}$ on $\mathbb{R}_{++}^2 = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$

(e). $f(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$, where $0 \leq \alpha \leq 1$, on $\mathbb{R}_{++}^2 = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$

5. Let $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ be an extended-valued convex function, i.e. for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\theta \in (0, 1)$,

$$f(\theta \mathbf{x} + \bar{\theta} \mathbf{y}) \leq \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{y})$$

- (a). For $\alpha \in (-\infty, +\infty]$, show $S_\alpha = \{\mathbf{x} : f(\mathbf{x}) < \alpha\}$ and $C_\alpha = \{\mathbf{x} : f(\mathbf{x}) \leq \alpha\}$ are convex using definition.
 (b). Show that the effective domain of f is convex.
 (c). Let $X \subset \mathbb{R}^n$ be convex and M the set of global minima of f over X , i.e.

$$M = \{\mathbf{x}^* \in X : f(\mathbf{x}^*) \leq f(\mathbf{x}), \forall \mathbf{x} \in X\}$$

Show M is convex. Hint: Show $M = X \cap C_\alpha$ for an appropriate α . You can assume $M \neq \emptyset$.

6. Suppose $f : S \subset \mathbb{R} \rightarrow \mathbb{R}$ is convex.

- (a). Show that for distinct $a, b, c \in S$ and $a < b$,

$$\frac{f(a) - f(c)}{a - c} \leq \frac{f(b) - f(c)}{b - c}$$

Hint: We already proved the case $c < a < b$ in class. You need to show the other cases, i.e. $a < c < b$ and $a < b < c$.

- (b). Show that if x is an interior point of S , the left and right derivatives of f at x

$$D_+ f(x) = \lim_{t \downarrow 0} \frac{f(x+t) - f(x)}{t},$$

$$D_- f(x) = \lim_{t \uparrow 0} \frac{f(x+t) - f(x)}{t} = \lim_{t \downarrow 0} \frac{f(x-t) - f(x)}{-t},$$

exist (as real numbers, not $\pm\infty$). Hint: prove the existence of the limit using the monotonicity of (a).

- (c). Define $g : [0, 1] \rightarrow \mathbb{R}$ by

$$g(x) = \begin{cases} 1, & \text{if } x = 0 \\ 0, & \text{if } x \in (0, 1] \end{cases}$$

Is g convex over $[0, 1]$? Does $D_+ f(0)$ exist? Does this contradict (b)?